Optimizing Athletes' Recovery and Performance: A Review of Vibration Therapy, Compression Garments, and Massage

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Abstract

It is evident that vibration therapy, massage, and compression garments are increasingly common recovery modalities among amateur and elite athletes. We performed a literature review via PubMed examining the effect of these modalities on measures of sports performance and recovery. We identified 29 eligible articles for vibration therapy, 31 eligible articles for compression garments, and 25 eligible articles for massage. Our review revealed varying degrees of support for each of the recovery modalities. However, our review also highlighted a need for standardized protocols in future studies to better understand how the modalities help athletes recover as to define optimal duration and frequency for maximal effects. Overall, all modalities reviewed seem safe to use and likely would not hinder sports performance or prolong recovery. It is a matter of how much benefit an athlete can gain when recovering from training.

Introduction

The continual record-breaking performances in athletics are a byproduct of advancements in training, recovery, and evolution of technology. It is becoming increasingly evident how crucial recovery contributes to these feats. In fact, according to PubMed, articles with the keyword *sports recovery* have grown from single digits in 1974 to 2309 in 2022. This article examines how much evidence is there to support the various recovery regimen that athletes incorporate in their training.

There are many metrics that an athlete can utilize to track their recovery. These can be as simple as monitoring pain, muscle stiffness, and delayed onset muscle soreness (DOMS). DOMS is widely considered a mild form of muscle injury that can compromise performance and increase chance of injury (1). Pressure pain threshold (amount of pressure an individual can withstand before feeling pain) can be used as a proxy for an athlete's recovery since it represents sensitivity of muscle to pressure (2).

In addition to these subjective measures, there are several objective metrics that are used as markers of recovery, including running or cycling times as well as strength, jump height, and range of motion.

Finally, there are biological and molecular markers of injury and recovery. These metrics include muscle swelling, muscle stiffness, muscle enzymes, blood lactate, inflammatory markers, and growth factors. Training results in microtearing of skeletal muscle, which results in spillage of enzymes such as lactate, creatine kinase, and myoglobin into the circulating blood (3). It is important to note that many of these lab values do not have established reference ranges and can range widely from one athlete to another, making interpretation of these values problematic (3). Therefore, trending these markers over time is often employed in these studies (4).

In this review article, we focus on three of the most popular recovery modalities: vibration therapy, compression garments, and massage, to determine their effectiveness on recovery.

Introduction of Recovery Modalities

Vibration Therapy

Whole body vibration is a recovery modality that has been proposed as an alternative to traditional methods such as massage, cryotherapy, and stretching. An individual stands, sits, or lies on a vibrating platform. It is theorized that vibration can help with recruitment of motor units and muscle fibers, which in turn accelerates recovery and decreases muscle soreness (4).

Compression Garments

Use of compression garments has shown promise in athletic recovery. The idea originated from clinical settings in which compression of calf muscle resulted in improved muscle oxygenation, decreased venous pooling, and edema (5). This was further studied in the athletic population and it was

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theorized that compression can help expedite recovery by removal of cellular waste, reduction in swelling and inflammation, and mechanically supporting the injured muscle fibers to allow proper alignment (5).

Massage

Massage is a popular and traditional recovery modality among athletes. Through use of mechanical pressure, it is postulated that massage decreases inflammatory response by increasing local circulation, thus removing accumulated cellular waste and increasing delivery of nutrients directly to the site of fatigued and damaged muscle (6). Massage also is thought to alter neural excitability through use of direct mechanical pressure, which affects muscle tension and thus decreases pain. This is done by reducing the amount of the stress hormone cortisol and increasing serotonin and dopamine levels (6).

Methods

For this literature review, PubMed was utilized. Relevant keywords and phrases pertinent to the area of interest were searched. Given the interest of this article is to examine recent research, inclusion criteria were set for journal articles from the year 2000 to present. Articles not in English were excluded. The journal articles were then examined based on their title and abstract to ensure the following subjective and objective findings were included: sports performance, muscle stiffness, strength, jump, inflammation markers, creatine kinase, myoglobin, blood lactate, swelling, perceived pain, pressure pain threshold, and range of motion.

The database was searched from the earliest date available to May 1, 2022 (date of the search), using the search terms massage, vibration, Theragun, Hypervolt, soft tissue mobilization, percussion, compression garment, normatec, myalgia, muscle soreness, muscle damage, and myalgia.

The data from the articles were grouped as follows: subjective measurements (visual analogue scale, pressure pain threshold, and self-rating scales), performance (sports performance, strength, jump, and range of motion), biomarkers (inflammatory markers, growth factors, muscle enzymes, and blood lactate), and objective measurements (swelling and muscle stiffness). The selected articles were evaluated to determine if the recovery modality resulted in significant improvement ($P < 0.05$), not significant changes ($P > = 0.05$), or significantly worse ($P < 0.05$) for each metric. It is important to note that not all articles reported all the metrics described above. The numbers of articles reviewed are as follows (see Figs. 1–2):

-Vibration Therapy: 29 articles
-Compression Garments: 31 articles
-Massage: 25 articles

Results

Vibration Therapy

Subjective measurements

There were 22 articles identified that looked at muscle pain and/or DOMS following vibration therapy. Of those, 17 articles found significant pain/DOMS reduction after vibration therapy (1,2,7–21). These studies included both highly trained and nontrained subjects. The vast majority used whole body vibration as the intervention. There was only one article that used a local vibrating foam roller versus a nonvibrating foam roller (16). Of note, two articles evaluated DOMS after eccentric exercise in elite athletes (professional ice hockey and futsal) and demonstrated significant improvement in pain (16,21). Four articles did not find any significant improvement in pain (22–25). Only one article found increased perceived pain after vibration therapy (26).

Performance

Performance following vibration therapy was evaluated in 18 studies. Of the 15 articles looking at strength, there were six articles that found significant improvement in strength following an exercise challenge — maximal voluntary contraction of bicep (17), grip (27), and knee (8,15,18,21) after vibration therapy. Eight articles did not find any significant improvement in strength after vibration therapy (2,9–11,23,24,26,28). Of note, there was one article reporting worsening strength 24 h following an eccentric challenge after therapy (29).

There was one article that found significant improvement in countermovement jump height after vibration therapy (14). There also were two articles that found no significant difference in vertical jumps after vibration therapy (24,25). There were no articles that found significantly worse countermovement jump height after vibration therapy.

Edge et al. evaluated performance in a 3-km running time trial following a high-intensity interval session followed by whole body vibration (WBV) therapy versus no WBV therapy. They found no difference in time trial performance utilizing WBV as a recovery tool (30).

Four articles looking at range of motion found faster recovery to baseline range of motion after vibration therapy (2,9,11,14). There were no articles that showed a significant decreased range of motion after vibration therapy. Of note, there was one article that demonstrated significantly increased dorsiflexion of the foot (+5.4 degrees) after one single vibration treatment session, though this was not after an exercise protocol (28).

Biomarkers

Of the three articles looking at inflammatory markers such as C-reactive protein, histamine, neutrophils, lymphocytes, interleukin 1 beta (IL1β), interleukin 6 (IL-6), and interleukin 10 (IL-10), one article found significant reduction after vibration therapy (7). Two articles did not find any significant difference after vibration therapy (23,26). There were no articles that found increased inflammatory markers after vibration therapy.

Of the 12 articles looking at markers of muscle damage (creatinine kinase and myoglobin), seven articles found significant reduction after vibration therapy (8–12,18,31). Five articles did not find any significant difference after vibration therapy (2,19,21,23,30). There were no articles that found increased creatine kinase or myoglobin after vibration therapy.

Of the seven articles that looked at clearance of blood lactate levels following exercise with and without vibration therapy, four articles found significant reduction in blood lactate after vibration therapy (9,12,31,32). Three articles did not find any significant difference after vibration therapy (21,24,30). There were no articles that found increased blood lactate level after vibration therapy.

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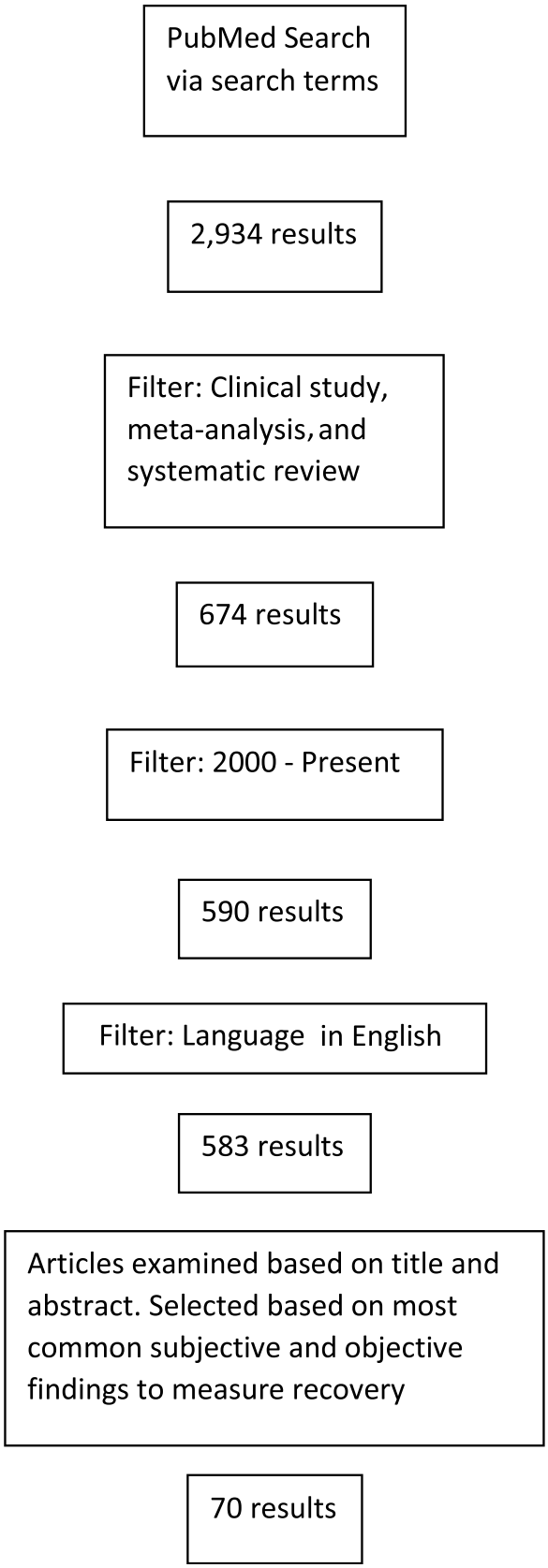

Figure 1: Details the systematic selection of studies examining recovery modalities. The initial dataset (n = 2,934) underwent progressive filtering based on study design, publication year, language, and relevance to clinical recovery metrics, yielding a final cohort of 70 articles for comprehensive analysis.

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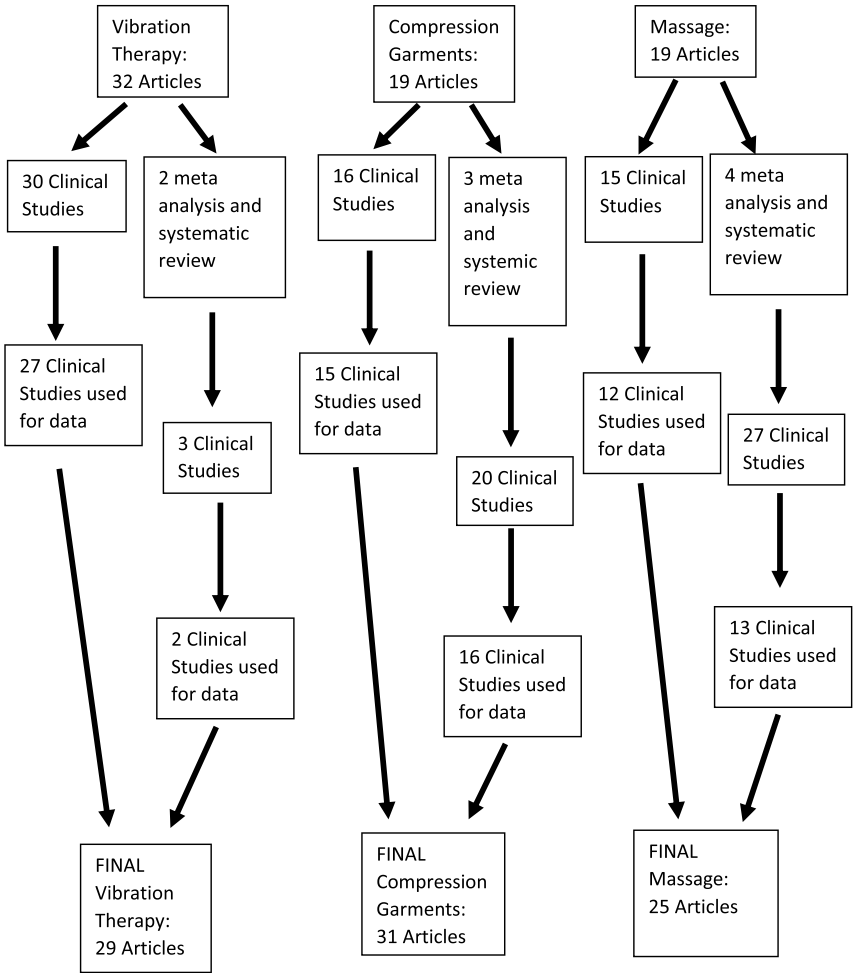

Figure 2: Categorizes the initial 70 articles from Figure 1 to the three modalities: vibration therapy, compression garments and massage. Then the articles in each modality is further categorized to clinical study or meta analysis/systematic review. The clinical studies are screened for relevance. The meta analysis/systematic review are screened and extracted for the original clinical studies. The final yield are as follows: 29 articles for vibration therapy, 31 articles for compression garments, and 25 articles for massage.

Objective Measures

There were three articles that evaluated muscle swelling (2,19,22). None of these found any significant difference in swelling after vibration therapy. There were no articles that found significant change in swelling after vibration therapy.

There was one article that investigated muscle stiffness after barbell curls when comparing control individuals to those who received vibration therapy (33). There was no change in muscle stiffness after vibration therapy. Interestingly, there were no articles that significantly improved or worsened muscle stiffness after vibration therapy.

Compression Garments

Subjective Measurements

There were 29 articles identified that looked at muscle pain or DOMS following use of compression garments. Of that, 16 articles found significant pain/DOMS reduction after wearing compression garments (5,34–48). These studies included both highly trained and nontrained subjects. The vast majority were examining lower extremity compression garments. There were three articles examining upper extremity compression garments (35,43,44) and three articles examining full body compression garments (39,42,47). Twelve articles did not find any significant improvement (49–60). Only one article found increased perceived pain after using compression garments (61).

Performance

Performance following compression garments was evaluated in 23 of the studies. Of the 12 articles looking at strength, there were eight articles that found significant improvement in strength following an exercise challenge — maximal voluntary contraction of knee (5,49,50,52), elbow (35,43,44), and chest (47) after wearing compression garments. Four articles did not find any significant improvement in strength — maximal voluntary contraction of knee (36,40), maximum squat and peak power of legs pushing a cart (41), and plantar flexor (45). There were no articles that found significantly decreased strength after wearing compression garments.

Of the eight articles looking at countermovement jump height, there were two articles that found significant improvement after wearing compression garments (5,52). Six articles found no significant difference after wearing compression garments (36,38,49,51,53,57). There were no articles that found

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worsening countermovement jump height after wearing compression garments.

Of the four articles looking at vertical jump, there was one article that found significant improvement after wearing compression garments (34). Three articles found no significant difference after wearing compression garments (5,38,56). There were no articles that found worsening vertical jump height after wearing compression garments.

Of the 11 articles looking at sports performance, there was one article that showed improved timed performance of line drill over the course of a basketball tournament after wearing compression garments (34). There were eight articles that revealed no significant difference in timed performance after wearing compression garments. The sports performances included sprints (40,42,49), drills (38,41,46,51,62), throwing distance, and accuracy (42). There were two articles that revealed decreased performance on sprints after wearing compression garments (34,57).

Of the five articles looking at range of motion, there were two articles that found faster recovery to baseline range of motion after wearing compression garments (43,44). Three articles did not find any significant difference after wearing compression garments (54,55,57). There were no articles that found significant decrease in range of motion after wearing compression garments.

Biomarkers

Of the eight articles looking at inflammatory markers and growth factors including C-reactive protein (CRP), tumor necrosis factor-alpha (TNF-alpha), aspartate transaminase (AST), glutamic oxalacetic transaminase (GOT), glutamic-pyruvic transaminase (GPT), gamma glutamyl transpeptidase (GGT), insulin-like growth factor 1 (IGF-1), free testosterone, cortisol, interleukin 6 (IL-6), interleukin 1 receptor (IL-1) antagonist, and interleukin-1ra, all eight found no significant difference after wearing compression garments (35,37,39,40,43,47,52,60). There were no articles indicating a significant improvement or worsening in inflammatory markers after wearing compression garments.

Of the 17 articles looking at markers of muscle damage (creatine kinase (CK) and myoglobin), five articles found significant reduction after wearing compression garments (36,42–44,49). Twelve articles demonstrated no significant difference after wearing compression garments (5,35,37,39–41,47,51–53,56,57). There were no articles that found increased CK or myoglobin after wearing compression garments.

Of the nine articles looking at reduction of blood lactate following exercise with and without compression garments, there was one article that found a significant reduction of blood lactate after wearing compression garments (63). Eight articles did not find any significant difference in blood lactate after wearing compression garments (37,39–43,47,51). There were no articles indicating significant increase in blood lactate after wearing compression garments.

Objective Measures

Of the 13 articles that evaluated muscle swelling, there were eight articles that found significant reduction after wearing compression garments (37,43,44,47,49,57,59,61). Five articles did not find any significant difference after wearing compression garments (34,51,54–56). There were no articles showing significant increase of swelling after wearing compression garments.

There was one article that revealed significantly less muscle stiffness after wearing compression garments (55). There were no articles that found no difference in muscle stiffness after compression garments. There were no articles that found significantly worse muscle stiffness after wearing compression garments.

Massage

Subjective Measurements

There were 19 articles identified that looked at muscle pain and/or DOMS following massage therapy. Seventeen articles found significant reduction in pain/DOMS after massage therapy (9,64–79). Two articles did not find any significant improvement (80,81). There were no articles that found significant worsening of perceived pain/pressure pain threshold after massage.

Performance

Performance following massage therapy was evaluated in 17 studies. Of the 10 articles looking at strength, there were five articles that found significant improvement in strength following an exercise challenge — maximal voluntary contraction of knee (66,72,74,82) and elbow (9) after massage therapy. Five articles did not find any significant improvement in strength — maximal voluntary contraction of thumb adductor (83), hamstrings (64), elbow (65), quadricep (81), and punching force boxing ergometer (77). There were no articles that found worsened strength after massage.

Of the two articles evaluating countermovement jump height, both articles found significant improvement after massage therapy (66,68). There were no articles that found no change or significant worsening of counter movement jump height after massage therapy.

There was one article that found significant improvement in squat jump height after massage therapy (66). There were no articles that found no change or significant worsening of squat jump height after massage therapy.

Of the four articles evaluating vertical jump height, there were two articles that found significant improvement after massage (72,75). One article found no significant difference in vertical jump height after massage (73). Another article found significant worsening in vertical jump height after massage (74).

There were six articles looking at sports performance, which included sprints, drills, high-intensity cycling, and swimming. Three articles revealed significant improvement in sports performance — sprint (68), cycling (84), and crawl swimming (85) — after massage therapy. Three articles did not find any significant difference in sports performance — 400-m run (71), agility (72), and sprint (73) — after massage therapy. There were no articles that found significant worsening in sports performance after massage therapy.

Of the four articles looking at range of motion, there were two articles that found faster recovery to baseline range of motion after massage (9,86). Two articles did not find any significant difference after massage therapy (64,65). There were no articles that found significant decreased range of motion after massage therapy.

Biomarkers

Of the two articles looking at inflammatory markers, there was one article that found significant reduction after massage therapy (87). The other article did not find any significant change after massage therapy (64). There were no articles that found increased inflammatory markers after massage.

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All three articles looking at muscle damage found significant reduction of CK/myoglobin after massage (9,65,72). There were no articles that found increase or no significant difference in CK/myoglobin after massage.

Of the five articles evaluating blood lactate, two articles found significant reduction after massage (84,85). Three articles found no significant difference after massage therapy (76,77,87). There were no articles that found significant increase in blood lactate after massage.

Objective Measures

There was one article that evaluated muscle swelling. One article found significant reduction after massage (65). There were no articles that found increased or no difference in swelling after massage.

Of the two articles looking at muscle stiffness, one article revealed significantly less muscle stiffness after massage therapy (76). The other article revealed no significant difference after massage therapy (80). There was no article that found increased muscle stiffness after massage therapy.

Discussion

Vibration Therapy

In our review, 17 out of 22 articles found perceived pain was significantly reduced after vibration therapy. Based on our review, a possible explanation is that vibration helps decrease exercise-induced muscle damage as shown by many articles with significant reduction of CK and myoglobin (8–12,18,31) and the increased clearance of blood lactate (9,12,31,32). These results are congruent with meta-analysis by Lu et al. (88). It is important to point out that peak pain relief and most significant decrease in CK occurs at 48 h after exercise (88). This coincides with inflammatory markers peaking at 48 h that provoke nerve endings resulting in perceived soreness (89). It is possible that vibration therapy creates better local circulation to remove cellular waste resulting in an improved local environment for recovery (90). These results align with the idea that vibration therapy helps with recruitment of muscle fibers to accelerate recovery and reduce muscle soreness (4). It appears that athletes were less stiff as four of four articles showed range of motion significantly improved after vibration therapy compared to control (2,9,11,14). However, although vibration therapy has positive effects on these metrics, it does not seem to translate into performance improvements. Of the 15 articles, only six articles found improved strength after vibration therapy (8,15,17,18,21,27). In addition, there is limited evidence of better performance with timed run and jump height after vibration therapy. It is worth noting that the duration of vibration therapy can affect strength and performance. According to Cardinale and Bosco, several minutes of vibration therapy can enhance strength, whereas longer duration can reduce strength possibly because of sensitization of muscle spindles or vibration activating inhibitory feedback (91). Despite its shortcoming in translating to performance metrics, vibration therapy does show promise as a recovery tool overall.

While our article focuses on review of postexercise vibration therapy, there is evidence that preexercise vibration therapy also can decrease exercise-induced muscle damage (92). According to Kim et al., preexercise vibration therapy was more effective than postexercise vibration therapy (12). Magoffin et al. supports this with preexercise vibration therapy most effective at 24 h compared to postexercise vibration therapy most effective at 48 h (15). The proposed mechanism is that preexercise vibration therapy enhances recruitment of muscle fibers, which decreases contractile stress on each unit resulting in reduced exercise-induced muscle damage (15). More research is needed to compare these differences.

Compression Garments

Use of compression garments is another alternative recovery modality that has shown potential. However, on further review, the aggregate data appear to suggest that this recovery modality falls short on its promise. The idea is that external pressure from the compression garments restricts the area available for swelling resulting in less inflammatory response and thus attenuates pain (93). Although there were many articles that found significant reduction in perceived pain after compression garment treatment — 16 out of 29 articles (5,34–48) — the other measurements do not support this theory. Decreased swelling does not seem to be consistently supported by the evidence. There were eight articles that found significant reduction of swelling after compression garment treatment (37,43,44,47,49,57,59,61). However, there also were five articles that were not able to find any significant difference in swelling after compression garment treatment (34,51,54–56). Further, there were eight articles that measured inflammatory markers and growth factors, and none of them were able to find a significant difference after compression garment treatment (35,37,39,40,43,47,52,60). This was the same with blood lactate with eight out of nine articles that were not able to find a significant difference in blood lactate after compression garment use (37,39–43,51). There are also claims of expedited removal of cellular waste to attenuate pain (5). However, this also does not seem to be supported by the current evidence as 12 of 17 articles showed no significant difference in CK or myoglobin after compression garment treatment (5,35,37,39–41,47,51,52,56,57). Lastly, there does not seem to be any performance or range of motion benefit with compression garments, though significant reduction in perceived pain may be of benefit to the athlete to return to training sooner. Overall, the evidence is weak to support compression garments as a recovery modality of choice.

Massage

Though only having equivocal evidence, massage is still a popular recovery modality among athletes. There are mixed results for massage's effects on recovery. It appears evident that massage reduces perceived pain and attenuates DOMS with 17 out of 19 studies reporting significant reductions in pain after treatment (9,64–79). According to Davis et al., massage reduced pain and attenuated DOMS by 13% (94). This is further supported by Guo et al. reporting that massage is more efficacious at 48 and 72 h postexercise compared to 24 h postexercise (90). Moreover, Poppendieck et al. revealed that massage for just 5–12 minutes appears to be sufficient to maximize the effects from massage (95). It also is important to point out that massage appears to have a larger effect in untrained subjects and more benefit for resistance type exercises (95). While there is strong evidence to support massage as a method to reduce perceived pain and attenuate DOMS, the scientific understanding of how massage does this is lacking.

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Although Crane et al. found significant dampening in the expression of inflammatory cytokines, there is an insufficient amount of evidence from our data review to elucidate further as only two studies total were found and only one study showed significant reduction in inflammatory markers after massage treatment compared to control (87). The findings from Crane et al. need further investigation to elucidate the effect of massage on inflammatory markers. Furthermore, there is minimal evidence to support the claims of increasing local circulation, removing cellular waste from exercise, and increasing the delivery of nutrients to sites of exercise-induced muscle damage. Although there is some evidence to support significant reduction of CK (9,65,72), the other factors such as inflammatory markers, blood lactate, and swelling have insufficient evidence to support such a claim (9,64,72,90). This concept will need further investigation.

However, massage does show promise. Lee et al. demonstrated the ability of massage to reduce motoneuron excitability, which is believed to affect muscle tension, and may explain the reduced perceived pain and attenuated DOMS (96). In addition, Field et al. and Rapaport et al. demonstrated massage's ability to reduce the stress hormone cortisol and increase serotonin and dopamine (97,98). Although it is unclear how these hormones attenuated DOMS, it is evident that massage has the ability to alter an individual's mindset, which can impact perceived pain intensity (96).

Limitations

It is important to point out the high heterogeneity of the studies included in this review. Firstly, there were major differences in each study design. The participants ranged in athletic ability levels from sedentary to novice to amateur and professional athletes. The sample size of each clinical study ranged as small as 8 subjects to the largest 220 subjects. Controls for the studies varied widely, some using separate control groups, and others with crossover design or using contralateral limb as control. Furthermore, the studies employed varying exercise protocols to induce muscle damage. Some studies used resistance exercises, whereas others had subjects participate in their sport. The resistance exercises varied in types of exercises performed, number of repetitions, and muscle groups of interest. The treatment protocols also were heterogeneous. The studies differed in duration and frequency of the treatment. Moreover, individual recovery modalities used had heterogeneous protocols. Vibration therapy studies had varying frequency (hertz) settings. Compression garments had varying pressure (mm Hg). Massage therapy was heterogeneous in techniques. Also, there were varying levels of expertise performing the massage, ranging from physical therapists to licensed massage therapists. In addition, there are no longitudinal studies investigating how the recovery modalities can be beneficial for athletes over the long term. These limitations provide a direction for future research to standardize protocols and approach to better understand these recovery modalities. Finally, although most outcomes reviewed had studies demonstrating both positive and negative results, it is important to note that data like these are subject to potential publication bias thus skewing results toward demonstrating positive findings.

Conclusion

As athletes, professional and amateur, look to improve their recovery to optimize their performance, it is important to continue to better understand the utility of commonly used recovery modalities: vibration therapy, compression garments, and massage therapy. This review revealed that there is some evidence supporting as well as refuting the use of each of the recovery modalities. In order to better understand their utility in sport, there needs to be more standardized protocols and approaches in future studies. Finally, it is important to note that the modalities reviewed in this article seem safe to use and likely would not hinder sports performance or prolong recovery. It is a matter of how much benefit an athlete can gain when recovering from training.

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